MANIPAL ACADEMY OF HIGHER EDUCATION
THIRD SEMESTER B.TECH. DEGREE EXAMINATIONS - NOVEMBER 2024
SUBJECT: CS 2101 - DATA STRUCTURES

Exam Date & Time: 25-Nov-2024 09:30 AM - 12:30 PM
Branch: Computer Science & Engineering CSE
Marks: 50 | Duration: 180 mins.

1A [5 Marks]
Construct an AVL tree for keys: 15, 20, 24, 10, 13, 7, 30, 36, 25. Show rotations.

1B [3 Marks]
Circular Queue implementation using fixed-size array with enqueue and dequeue bounds.

2A [5 Marks]
Dijkstra single-source shortest path algorithm with priority queue implementation.

2B [4 Marks]
QuickSort partition algorithm with random pivot selection and average-case analysis.

3A [5 Marks]
Binary Search Tree deletion: handle leaf node, single child, and two children cases.

3B [4 Marks]
Kruskal Minimum Spanning Tree algorithm using Disjoint Set Union DSU .

4A [5 Marks]
Collision resolution in Hash Tables: Quadratic Probing vs Separate Chaining comparison.

4B [4 Marks]
Topological Sort on Directed Acyclic Graphs DAG using Kahn algorithm in-degree .

CS 2101: DATA STRUCTURES [END OF PAPER]